The calibration system and its scaling method of digital dust instrument

Abstract

The invention discloses a kind of calibration system of digital dust instrument and its scaling method, the calibration system includes the calibration system of cutting head and the calibration system of dust instrument mass concentration；The calibrating device of cutting head includes compressed air treatment device, the compressed air treatment device connects monodisperse particle generator, monodisperse particle generator connects uniform blending bin, uniform blending bin connects dust arrester, isokinetic sampling's head respectively, two isokinetic sampling's heads connect standard cutting head and tested cutting head respectively, tested cutting head connects the first filter membrane, and standard cutting head connects the second filter membrane, and the first filter membrane and the second filter membrane connect sampling apparatus and automatic weighing device simultaneously；The calibration system of dust instrument mass concentration includes compressed air treatment device, dust generator, uniform blending bin, dust arrester, the first standard cutting head, the second standard cutting head, digital dust instrument, the 3rd filter membrane and sampling apparatus, the 3rd filter membrane connects automatic weighing device.The present invention makes the calibration operation of dust instrument obtain standardized management, improves the degree of accuracy of apparatus measures.

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Description

The calibration system and its scaling method of digital dust instrument

Technical field

The present invention relates to dust measurement field, and in particular to the calibration system and its scaling method of a kind of digital dust instrument.

Background technology

Digital dust instrument is to include respirable dust instrument, explosion-proof type dust for measuring the instrument of various bug dust concentrations Instrument, micro computer dust instrument, it is widely used in IAQ detection, workplace air quality testing and open-air matter Amount detection.The power and cutting efficiency accuracy of dust instrument mass concentration indicating value are to influence the important parameter of measurement result.At present Situation be that not only method is different for the Quality Control Technology of producer, and very lack of standardization, some simply flow parameter is carried out Evaluation, and China lacks technical equipment and technical specification to the magnitude tracing of the instrument, also results in measurement data Confusion, comparability is poor, therefore sets up the calibration system and its scaling method of a set of digital dust instrument, improves the amount of the quasi-instrument Value is traced to the source transmission system, specification instrument performance evaluation so that the magnitude tracing of digital dust instrument works to obtain standardized management.

The content of the invention

The brief overview on the present invention is given below, to provide the basic reason on certain aspects of the invention Solution.It should be appreciated that this general introduction is not the exhaustive general introduction on the present invention.It is not intended to determine the key of the present invention Or pith, nor is it intended to limit the scope of the present invention.Its purpose only provides some concepts in simplified form, with This is as the preamble in greater detail discussed later.

The purpose of the embodiment of the present invention be for existing dust instrument response curve and the degree of accuracy vary the defects of, The calibration system and its scaling method of a kind of digital dust instrument are provided, to examine and determine the dust instrument of different manufacturers and different user, So that the detection work of dust instrument obtains standardized management.

To achieve these goals, the present invention adopts the technical scheme below：

A kind of calibration system of digital dust instrument, including the demarcation system of the calibration system of cutting head and dust instrument mass concentration System；

The calibration system of the cutting head includes compressed air treatment device, and the compressed air treatment device connects single-point Shot electronic generator, the monodisperse particle generator connect uniform blending bin, and the uniformly blending bin connects dedusting dress respectively Put, standard cutting head and tested cutting head, the tested cutting head connects the first filter membrane, the standard cutting head connects second Film, first filter membrane and the second filter membrane connect sampling apparatus and automatic weighing device simultaneously；

The calibration system of the dust instrument mass concentration includes compressed air treatment device, the compressed air treatment device Connect dust generator, the dust generator connects uniform blending bin, and the uniform blending bin connects dust arrester, the first mark Respectively Quasi- cutting head and the second standard cutting head, the first standard cutting head connect digital dust instrument, and second standard is cut Cut head and the second standard cutting head, the first standard cutting head connect digital dust instrument, and second standard is cut Head is sequentially connected the 3rd filter membrane and sampling apparatus, the 3rd filter membrane connects automatic weighing device.

The present invention also provides a kind of scaling method of the calibration system using above-mentioned digital dust instrument, includes the mark of cutting head Determine the scaling method of method and dust instrument mass concentration：

(1) scaling method of cutting head：

After compressed air is passed through monodisperse particle generator, powder sample caused by monodisperse particle generator enters uniform Blending bin, after Xelminator eliminates electrostatic, it is well mixed with compressed air in uniform blending bin, is adopted by isokinetic sampling's head The powder sample of collection out respectively enters standard cutting head and tested cutting head；By the two-way adjustable gas flowmeter of sampling Apparatus Same traffic is adjusted into, under identical acquisition time, first filter membrane and the second filter membrane are weighed using automatic weighing device；The ratio of granular mass collected by first filter membrane and the second filter membrane is the cutting efficiency of the standard cutting head；Make respectively The entrance cutting efficiency of standard cutting head, draws cutting curve under at least five kinds of particle diameters；Correspondingly the particle diameter of 50% cutting efficiency is Cut point, judge whether the cut point and the nominal cut point of instrument are consistent；

(2) scaling method of digital dust instrument mass concentration：

After compressed air is passed through dust generator, enter uniform blending bin through powder sample caused by dust generator, pass through After Xelminator eliminates electrostatic, it is well mixed in uniform blending bin, by isokinetic sampling's head Powder sample caused by dust generator, pass through After Xelminator eliminates electrostatic, it is well mixed in uniform blending bin, gathers out by isokinetic sampling's head Powder sample respectively enters standard cutting head：The flow of the adjustable gas flowmeter of sampling apparatus is adjusted to and Digital Dust to be measured The flow of instrument is identical, under identical acquisition time, reads the stable reading of digital dust instrument, meanwhile, utilize automatic weighing device pair 3rd filter membrane is weighed；The ratio of quality that the 3rd filter membrane weighs and sampling volume is calculated, bug dust concentration is obtained, judges Whether the bug dust concentration is consistent with the reading of digital dust instrument mass concentration；At least three chosen in the range ability of dust instrument Concentration value is contrasted.Compared with prior art, the beneficial effects of the invention are as follows：

The invention provides a kind of the calibration system of digital dust instrument and its scaling method, for different light scattering apparatus Device, the demarcation of mass concentration is carried out under unified, stable systematic medium, so as to examine and determine the dust instrument of different manufacturers and different user So that the detection operation of dust instrument is able to obtain standardized management, while improve the degree of accuracy of apparatus measures.The side of the present invention Method has NIST (National Institute of Standards and Technology) tractability.

Brief description of the drawings

In order to illustrate more clearly about the embodiment of the present invention or technical scheme of the prior art, below will be to embodiment or existing There is the required accompanying drawing used in technology description to be briefly described, it should be apparent that, the accompanying drawings in the following description are only Some embodiments of invention, for those of ordinary skill in the art, without having to pay creative labor, may be used also To obtain other accompanying drawings according to these accompanying drawings.

Fig. 1 is the schematic flow sheet of the calibration system of cutting head provided in an embodiment of the present invention；

Fig. 2 is the schematic flow sheet of the calibration system of dust instrument mass concentration provided in an embodiment of the present invention；

Fig. 3 is uniformly blending bin front view provided in an embodiment of the present invention；

Fig. 4 is cutting curve figure provided in an embodiment of the present invention.

Reference：

1-compressed air treatment devices；2-monodisperse particle generators；The uniform blending bins of 4-；5-dust arresters；6-standards Cutting head；7-is detected cutting head；The filter membranes of 8-first；9-sampling apparatuses；10-automatic weighing devices；11-dust generators；12-is adopted Sampling device；The filter membranes of 13-second；14-the first standard cutting head；15-the second standard cutting heads；The filter membranes of 16-the 3rd；17-numeral powder Dirt instrument；18-compressed air inlets；19-Xelminators；20-isokinetic sampling's heads；21-dust removal ports；22-casings；23-dust enters Mouthful.

Embodiment

To make the purpose, technical scheme and advantage of the embodiment of the present invention clearer, below in conjunction with the embodiment of the present invention In accompanying drawing, the technical scheme in the embodiment of the present invention is clearly and completely described, it is clear that described embodiment is Part of the embodiment of the present invention, rather than whole embodiments.Described in the accompanying drawing of the present invention in one accompanying drawing or a kind of embodiment Element and the element that can be shown in one or more other accompanying drawings or embodiment and feature and feature be combined.Should Pay attention to, for purposes of clarity, eliminated in accompanying drawing and explanation and is unrelated to the invention, known to persons of ordinary skill in the art Part and the expression and description of processing.Based on the embodiment in the present invention, those of ordinary skill in the art are not paying creative Property work under the premise of the every other embodiment that is obtained, belong to the scope of protection of the invention.

Referring to Fig. 1 and Fig. 2, a kind of calibration system of digital dust instrument, including the calibration system of cutting head and dust instrument matter Measure the calibration system of concentration；

The calibration system of the cutting head includes compressed air treatment device 1, compressed air treatment device 1 connects dispersing Particle generator 2, monodisperse particle generator 2 connect uniform blending bin 4, uniform blending bin 4 connects dust arrester 5, mark respectively Quasi- cutting head 6 and tested cutting head 7, tested cutting head 7 connects the first filter membrane 8, standard cutting head 6 connects the second filter membrane 13, the first filter membrane 8 and the second filter membrane 13 connect sampling apparatus 9 and automatic weighing device 10 simultaneously；

The calibration system of the dust instrument mass concentration includes compressed air treatment device 1, compressed air treatment device 1 connects dust generator 11, dust generator 11 connects uniform blending bin 4, uniform blending bin 4 connects dust arrester 5, first respectively Standard cutting head 14 and the second standard cutting head 15, the first standard cutting head 14 connects digital dust instrument 17, the second standard cutting head 14 connection digital dust instrument 17, the second standard cutting head 15 is sequentially connected the 3rd filter membrane 16 and sampling apparatus 12, the 3rd filter membrane 16 connection automatic weighing device 10.

Invention realizes the calibration operation of dust instrument obtain standardized management, improves the accuracy of apparatus measures measures.

In the present embodiment, referring to Fig. 3, uniform blending bin 4 includes casing 22, and the top of casing 22 is provided with dust inlet 17, the lower square box 22 of dirt entrance 17 is provided with Xelminator 19, dust inlet 17 is provided with compressed air entrance 18, Xelminator 19 Lower section is provided with isokinetic sampling head 20, and the bottom of casing 22 is provided with dust removal port 21.

Preferably, box volume about 150L, the circular cylinder to be bound for both ends with gap, diameter 35cm, height 150cm If There is an internal diameter 4 mm (inner diameter 4mm) dust inlet；One compressed air treatment device entrance (internal diameter 20mm)；Excel built in Xelminators (being distributed in " ten " font, arc of recess top 10cm, Xelminator long 10cm, wide 2cm)；The isokinetic sampling that four internal diameters are 10mm Head；One internal diameter diameter 10mm dust removal port.

In the present embodiment, the material of casing 22 is 304 stainless steel plate；The internal face of casing 22 is by processing of scraping, its surface roughness is less than 2.5 μm, and to prevent electrostatic and granule absorption, Xelminator 19 is source of the gas type Xelminator, in " ten " Font is placed on same level in casing 22.In the present embodiment, compressed air treatment device 1 be by oil-free silent air compressor, Cooling driers, stage high-efficiency filter and surge tank composition, dried and clean compressed air after treatment are stored in air accumulator.

Oil-free silent air compressor is stored in the air in air accumulator after being made.

In the present embodiment, standard cutting head 6 includes PM10 cutting heads and PM2.5 cutting heads, wherein the first standard cutting head 14 and second standard cutting head 15 can use PM10 cutting heads or PM2.5 cutting heads；First filter membrane 8, the second filter membrane 13 and the 3rd filter membrane 16 are 47mm teflon membrane filter.

PM (Particulate Matter, particulate matter), PM10 represent particulate air kinetic diameter as 10 microns, TSP (Total Suspended Particulate, also known as total suspended particles, overall suspended pellet).

In the present embodiment, automatic weighing device 10 is used to realize automatic sampling by manipulator and precision electronic balance and claimed Weight, also there is a atomic dispersion device constant temperature, antidetonation, filter membrane to weigh, Automatic change film and the function with wireless data transmission.

The automatic weighing device of the present invention is to realize to adopt automatically by manipulator and precision electronic balance (a ten thousandths) Sample, weigh, device contains Automated control software, realizes human-machine operation Claim with constant temperature (25 ± 0.5 DEG C), antidetonation, filter membrane Weighing The functions of automatic exchange filter film, data wireless transmission. The automatic weighing device (filter membrane balance (ten thousandths), Weighing the automatic exchange filter membrane housing, carries manipulator, the control of 2 filter membranes System and executing agency, precision electronic balance (ten thousandths) thereon, constant temperature and the interior work(with constant temperature of housing of box, base is installed below housing, the precision electronic balance of level of box is placed on base, makes dress of entirely weighing apparatus Put be fixed on function of seismic resistance, manipulator on housing, manipulator is used to capture filter membrane and be positioned in precision electronic balance to carry out Weigh, resulting quality, by carrying out computing with the system electronic balance inputted in computer, is obtained by wireless data transmission to computer To the concentration of dust.

In the present embodiment, sampling apparatus 9 is with two-way adjustable gas flowmeter Sampling pump；Sampling apparatus 12 is the sampling pump with single channel adjustable gas flowmeter.

In the present embodiment, dust arrester 5 is the external vacuum cleaner for removing dust on uniform blending bin bottom.

The present invention also provides a kind of scaling method of digital dust instrument, including the scaling method of cutting head and dust instrument quality The scaling method of concentration：

(1) scaling method of cutting head：

After compressed air is passed through monodisperse particle generator, powder sample caused by monodisperse particle generator enters uniform Blending bin, after Xelminator eliminates electrostatic, it is well mixed with compressed air in uniform blending bin, is adopted by isokinetic sampling's head The powder sample of collection out respectively enters standard cutting head and tested cutting head；By the two-way adjustable gas flowmeter of sampling apparatus Same traffic is adjusted into, under identical acquisition time, the first filter membrane and the second filter membrane are weighed using automatic weighing device；Calculate cutting efficiency；The entrance cutting efficiency of cutting head under at least five kinds of particle diameters is made respectively, draws cutting curve；It is corresponding The particle diameter of 50% cutting efficiency is cut point, judges whether the cut point and the nominal cut point of instrument are consistent；

(2) scaling method of dust instrument mass concentration：

After compressed air is passed through dust generator, enter uniform blending bin through powder sample caused by dust generator, pass through After Xelminator eliminates electrostatic, it is well mixed in uniform blending bin, by isokinetic sampling's head Powder sample respectively enters standard cutting head：The flow of the adjustable gas flowmeter of sampling apparatus is adjusted to and Digital Dust to be measured The flow of instrument is identical, under identical acquisition time, reads the reading of digital dust instrument, meanwhile, utilize automatic weighing device pair 3rd filter membrane is weighed；The ratio of quality that the 3rd filter membrane weighs and sampling volume is calculated, bug dust concentration is obtained, judges Whether the bug dust concentration is consistent with the reading of dust instrument mass concentration；At least three are chosen in the range ability of dust instrument Concentration value is contrasted.

The present invention also provides a kind of scaling method of cutting head and dust instrument mass concentration, including the scaling method of cutting head and dust instrument quality The scaling method of concentration：

(1) scaling method of cutting head：

After compressed air is passed through monodisperse particle generator, powder sample caused by monodisperse particle generator enters uniform Blending bin, after Xelminator eliminates electrostatic, it is well mixed with compressed air in uniform blending bin, is adopted by isokinetic sampling's head The powder sample of collection out respectively enters standard cutting head and tested cutting head；By the two-way adjustable gas flowmeter of sampling apparatus Same traffic is adjusted into, under identical acquisition time, filter membrane is weighed using automatic weighing device；Collected by two filter membranes Granular mass ratio for standard cutting head entrance cutting efficiency；Standard cutting head under at least five kinds of particle diameters is made respectively Entrance cutting efficiency, draw cutting curve；The particle diameter of corresponding 50% cutting efficiency is cut point, judges the cut point and instrument Whether the nominal cut point of device is consistent；

(2) scaling method of dust instrument mass concentration：

After compressed air is passed through dust generator, enter uniform blending bin through powder sample caused by dust generator, pass through After Xelminator eliminates electrostatic, it is well mixed in uniform blending bin with compressed air, gathers out by isokinetic sampling's head Powder sample respectively enters standard cutting head；The flow of the adjustable gas flowmeter of sampling apparatus is adjusted to and Digital Dust to be measured The flow of instrument is identical, under identical acquisition time, reads the stable reading of digital dust instrument, meanwhile, utilize automatic weighing device pair filter membrane is weighed；The ratio of quality that the filter membrane weighs and the standard sampling volume is multiplied by the sampling time by standard sample flow and obtained, and calculates quality that Filter membrane is weighed with the sampling volume ratio, obtains bug dust concentration and dust instrument mass concentration reading are consistent；At least three concentration values are chosen in the range ability of dust instrument to be contrasted.

In the present embodiment, the particle diameter of the powder sample at least selects following five kinds：5 μm, 8 μm, 10 μm, 12 μm and 15 μm.

The scaling method of the present invention is described as follows by specific embodiment：

The demarcation of inhalable digital dust instrument divides two parts：The demarcation of PM10 cutting heads and the demarcation of mass concentration. The scaling methods of PM10 cutting heads as shown in Figure 1, it is necessary at least five kinds of particle diameters single dispersing PSL particles (such as 5 μm, 8 μm, 10 μm, 12 μm, 15 μm etc.) To the dry powder sample of every kind of particle diameter, after compressed air 1 is passed through monodisperse particle generator 2 first, single dispersing Powder sample caused by particle generator 2 enters uniform blending bin 4, after Xelminator 3 eliminates electrostatic, in uniform blending bin 4 well mixed with compressed air 1, the powder sample for gathering out by isokinetic sampling's head 4 respectively enters standard cutting head 6 and by Examine cutting head 7；The two-way adjustable gas flowmeter of sampling apparatus 9 is adjusted into same traffic, utilized Automatic weighing device 10 to weigh the first filter membrane 8 and the second filter membrane 13；The ratio of granular mass collected by two filter membranes For the cutting efficiency of standard cutting head 6, show that cutting curve (as shown in Figure 4)；Referring to Fig. 4, abscissa is particle diameter, unit for μm, Ordinate is cutting efficiency.The particle diameter of corresponding 50% cutting efficiency is cut point, judges that the cut point is nominally cut with instrument Whether point is consistent (Instrument is PM10, nominally the nominal cut point of instrument is 10μm).The grain of corresponding 50% cutting efficiency Footpath is cut point, judge the cut point and the nominal cut point of instrument whether consistent.

The scaling method of digital dust instrument mass concentration is as shown in Figure 2.

After compressed air 1 is passed through dust generator 11, enter uniformly blending bin through powder sample caused by dust generator 11 Case 4, after Xelminator 3 eliminates electrostatic, it is well mixed with compressed air 1 in uniform blending bin 4, the powder sample is adopted by isokinetic sampling's head The powder sample of collection out respectively enters standard cutting head 6：By the flow of digital dust instrument 17 to be measured The flow of the adjustable gas flowmeter of sampling apparatus 12 is adjusted to and the flow of digital dust instrument 17 to be measured is identical, under identical acquisition time, reads the stable reading of digital dust instrument 17, meanwhile, The 3rd filter membrane 16 is weighed using automatic weighing device 10, the quality and sampling volume is multiplied by the sampling time by standard sample flow and obtained Ratio, obtains bug dust concentration, judges whether the bug dust concentration and digital dust instrument mass concentration reading are consistent；At least three concentration values are chosen in the range ability of digital dust instrument to be contrasted.

The measurement result of the present embodiment is as follows：

Finally it should be noted that：The above embodiments are merely illustrative of the technical solutions of the present invention, rather than its limitations；Although The present invention is described in detail with reference to the foregoing embodiments, it will be understood by those skilled in the art that：It still may be used to be modified to the technical scheme described in foregoing embodiments, or equivalent substitution is carried out to which part technical characteristic；And these modification or replace, do not make appropriate technical solution essence depart from various embodiments of the present invention technical scheme spirit and Scope.

Claims (7)

Hide Dependent

1. a kind of calibration system of digital dust instrument, it is characterised in that calibration system and dust instrument quality including cutting head are dense The calibration system of degree；

The calibration system of the cutting head includes compressed air treatment device, and the compressed air treatment device connects single dispersing grain Electronic generator, the monodisperse particle generator connect uniform blending bin, and the uniformly blending bin connects dust arrester, mark respectively Quasi-cutting head and tested cutting head, the tested cutting head connect the first filter membrane, and the standard cutting head connects the second filter membrane, institute State the first filter membrane and the second filter membrane while connect sampling apparatus and automatic weighing device；

The calibration system of the dust instrument mass concentration includes compressed air treatment device, the compressed air treatment device connection Dust generator, the dust generator connect uniform blending bin, the uniformly blending bin connects dust arrester, the first mark respectively Quasi- cutting head and the second standard cutting head, the first standard cutting head connect digital dust instrument, the second standard cutting head The 3rd filter membrane and sampling apparatus are sequentially connected, the 3rd filter membrane connects automatic weighing device；

The uniformly blending bin includes casing, and the casing top is provided with dust inlet, on the lower square box of dust inlet It is provided with compressed air inlet, Xelminator, institute are provided with by the casing between the dust inlet and the compressed air inlet State lower section Xelminator is provided with isokinetic sampling's head, the bottom half is provided with dust removal port；

The compressed air treatment device is made up of oil-free silent air compressor, cooling driers, stage high-effective filter and surge tank.

2. the calibration system of digital dust instrument according to claim 1, it is characterised in that the material of the casing is 304 Stainless steel plate；The internal face of the casing is less than 2.5 μm by processing of scraping, its surface roughness, and to prevent electrostatic and granule absorption, Xelminator is source of the gas type Xelminator, same level in casing is placed in " ten " font.

3. the calibration system of digital dust instrument according to claim 1, it is characterised in that the standard cutting head includes PM10 cutting heads and PM2.5 cutting heads；First filter membrane, the second filter membrane and the 3rd filter membrane are 47mm polytetrafluoroethylene (PTFE) filter Film.

4. the calibration system of the digital dust instrument according to claim any one of 1-3, it is characterised in that described be to weigh automatically Device, which is used to realize by manipulator and precision electronic balance, to be sampled and weighs automatically, and the automatic weighing device also has perseverance Temperature, antidetonation, filter membrane are weighed, Automatic change film and the function being wirelessly transferred with data.

5. the calibration system of digital dust instrument according to claim 4, it is characterised in that the sampling apparatus is with double Road adjustable gas flowmeter or the sampling pump with single channel adjustable gas flowmeter.

6. the calibration system of digital dust instrument according to claim 4, it is characterised in that the dust arrester is to be arranged on the external vacuum cleaner of uniformly blending bin bottom.

7. a kind of usage right requires the scaling method of any one of the 1-6 calibration system of the digital dust instrument, it is characterised in that The scaling method of scaling method and dust instrument mass concentration including cutting head：

(1) scaling method of cutting head：

After compressed air is passed through monodisperse particle generator, powder sample caused by monodisperse particle generator enters uniformly mixing Case, after Xelminator eliminates electrostatic, it is well mixed in air with compressed air in uniform blending bin, is gathered out by isokinetic sampling's head The powder sample respectively enters standard cutting head and tested cutting head；The two-way adjustable gas flowmeter of sampling apparatus is adjusted to Into same traffic, under identical acquisition time, the first filter membrane and the second filter membrane are weighed using automatic weighing device；The ratio of granular mass collected by filter membrane and the second filter membrane is the cutting efficiency of standard cutting head；Make respectively The entrance cutting efficiency of standard cutting head, draws cutting curve under at least 5 kinds of particle diameters；The particle diameter of corresponding 50% cutting efficiency is to cut Point, judge whether the cut point and the nominal cut point of instrument are consistent；

(2) scaling method of dust instrument mass concentration：

After compressed air is passed through dust generator, enter uniform blending bin through powder sample caused by dust generator, through Xelminator eliminate after electrostatic, it is well mixed in uniform blending bin with compressed air, the dust for gathering out by isokinetic sampling's head Sample respectively enters standard cutting head；The flow of the adjustable gas flowmeter of sampling apparatus is adjusted to and digital dust instrument to be measured Flow is identical, under identical acquisition time, reads the stable reading of digital dust instrument, meanwhile, utilize automatic weighing device is to the 3rd Filter membrane is weighed；The ratio of quality that the 3rd filter membrane weighs and sampling volume is calculated, bug dust concentration is obtained, judges whether the bug dust concentration is consistent with the reading of dust instrument mass concentration；At least three concentration are chosen in the range ability of dust instrument Value is contrasted.

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CN104111217A *	2014-06-18	2014-10-22	瑞曼博(北京)环保科技有限公司	Dust Calibration System
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Priority And Related Applications